

Teleparallel Gauss–Bonnet gravity beyond flat FLRW cosmology

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We investigate the cosmological dynamics of teleparallel Gauss–Bonnet gravity in spatially curved FLRW spacetimes. We construct a class of spatially curved tetrads that provides a unified description of open, flat, and closed FLRW geometries and allows the teleparallel torsion scalar and Gauss–Bonnet invariant to be expressed directly in terms of the spatial curvature parameter. Using a scalar-field representation of the theory, we derive an autonomous dynamical system that remains well-defined in both expanding and contracting cosmological phases. We analyze the critical points and their stability for flat and spatially curved backgrounds, revealing a rich phase-space structure with distinct curvature-dependent asymptotic behaviors. In particular, the curved system admits a stable attractor on the flat submanifold, where the curvature contribution is dynamically suppressed, whereas other asymptotic solutions allow spatial curvature to persist.

I. INTRODUCTION

General Relativity successfully describes gravity in the weak-field regime and satisfies local tests, but its cosmological application faces unresolved issues, particularly the nature of the dark sector. This motivates modified gravity theories as possible explanations for cosmic acceleration [1–3]. Among them, modified teleparallel gravity reformulates gravity in terms of torsion rather than curvature, building on the teleparallel equivalent of general relativity (TEGR) [4–11]. Its extensions promote the torsion scalar and related invariants to general functions [12, 13].

The original formulation of TEGR suffers from loss of local Lorentz invariance due to setting the spin connection to zero and treating the tetrad as the sole dynamical variable. Although the resulting boundary term does not affect TEGR field equations, it becomes dynamically relevant in modified teleparallel theories, making locally Lorentz-related tetrads inequivalent [14, 15]. This led to the distinction between “good” and “bad” tetrads [16, 17]. A Lorentz-covariant formulation was later established by treating both the tetrad and teleparallel spin connection as fundamental variables [11, 18–20]. In modified theories, variation with respect to the spin connection yields consistency conditions related to the antisymmetric part of field equations obtained via variation with respect to tetrads [21, 22]. Nevertheless, finding the appropriate tetrad–spin-connection pair for general geometries remains technically challenging [18, 23].

This difficulty is particularly important for spatially curved FLRW cosmology. Unlike curvature-based and symmetric teleparallel theories, modified teleparallel gravity lacks a known good tetrad yielding field equations directly parameterized by the spatial curvature k . Consequently, the cases $k = +1$ and $k = -1$ are usually treated separately, with a general structure inferred afterward [24–28]. This approach is cumbersome and may fail for non-minimal TEGR extensions [29]. Beyond its technical inconvenience, this limitation obscures the dynamical role of spatial curvature in teleparallel cosmology [30, 31].

In this work, we address this problem by constructing a class of tetrads for FLRW spacetimes that yields all relevant teleparallel quantities directly in terms of the curvature parameter k , thereby providing a unified and consistent description valid for open ($k = -1$), flat ($k = 0$) and closed ($k = +1$) cosmologies. The construction is achieved by applying a suitable local Lorentz rotation to the diagonal tetrad, resulting in configurations that preserve homogeneity and isotropy at the level of the torsion invariants. As an application of this framework, we investigate the dynamics of the curved tetrads in the context of teleparallel Gauss–Bonnet theory. Gauss–Bonnet modifications of gravity generalize TEGR by promoting the teleparallel Gauss–Bonnet scalar T_G to a nontrivial function in the gravitational action. The $\mathcal{F}(T, T_G)$ framework and field equations were introduced in Ref. [20], followed by the background cosmological equations and model studies in Ref. [32]. Its dynamical properties were subsequently investigated in Ref. [33], while early-universe scenarios and observational constraints were explored in Refs. [34–39]. Gravitational waves in this framework propagate at the speed of light [40], consistent with multimessenger constraints [41, 42], unlike the curvature-based analog [43]. The perturbative sector and stability properties were developed in Ref. [44], while the evolution of matter density perturbations was studied in Ref. [45]. These studies have established several interesting cosmological and perturbative properties of the theory. However, a unified treatment of spatially curved FLRW backgrounds remains lacking. By reformulating the theory in terms of an effective scalar-field representation,

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we derive a dynamical system that captures the evolution of both flat and curved cosmologies in a unified phase-space description. This allows us to investigate the curvature-dependent phase-space structure and identify distinct asymptotic behaviors.

The paper is organized as follows. In Sec. II, we present the technical foundations of teleparallel gravity, culminating in the introduction of the teleparallel Gauss–Bonnet Lagrangian. In Sec. III, we construct curvature-dependent tetrads for FLRW cosmologies, which are subsequently employed to derive the invariants leading to Friedmann equations of the theory. In Sec. IV, we reformulate the Lagrangian in terms of an equivalent non-canonical scalar field representation and derive the corresponding evolution equations, which are then analyzed using dynamical systems techniques in Sec. V. Finally, in Sec. VI, we summarize the main results obtained in this work, emphasize the key implications of the formalism under consideration and outline potential directions for future research.

II. TELEPARALLEL GAUSS-BONNET GRAVITY

In gravitational theory, the fundamental structure of interest is the spacetime manifold $\mathcal{M}$, modeled as a four-dimensional, connected, Hausdorff, and paracompact C^∞ manifold [46]. Endowed with a locally Lorentzian metric $g = \{g_{\rho\sigma}\}$, this manifold admits absolute parallelism through four linearly independent parallel vector fields $\{e_a\} = \{e_a^\rho\}$ at each point. These “vierbeins” (or tetrads) define the teleparallel connection as,

$$\Gamma^\tau_{\rho\sigma} := e_a^\tau \partial_\sigma e^a_\rho + e_a^\tau \omega^a_{b\sigma} e^b_\rho, \quad (1)$$

where the dual set $\{e^a\} = \{e^a_\rho\}$ also consists of parallel vector fields. The spin connection components $\omega^a_{b\sigma}$ define parallel transport and encode inertial effects. In this framework, Latin indices $(a, b, \dots)$ represent local Lorentz indices, while Greek indices $(\rho, \sigma, \dots)$ represent global spacetime indices. Vierbeins satisfy an orthogonality condition relative to the metric tensor, $g(e_a, e_b) = \eta_{ab}$, where $\eta_{ab} = \text{diag}(-1, 1, 1, 1)$. This establishes the metric reconstruction,

$$g_{\rho\sigma} = \eta_{ab} e^a_\rho e^b_\sigma. \quad (2)$$

Lorentz indices are manipulated via the Minkowski metric η_{ab} and spacetime components are obtained by contracting Lorentz indices with the tetrads or their duals. The vierbeins associated with a specific metric (2) are not unique. To preserve covariance, the spin connection must be selected to compensate for the chosen tetrad. Since direct adoption of Minkowskian geometry can introduce spurious inertial effects, an inertial spin connection is utilized. This connection is flat i.e. satisfying

$$\partial_{[\rho} \omega^a_{b|\sigma]} + \omega^a_{c[\rho} \omega^c_{b|\sigma]} = 0. \quad (3)$$

To ensure consistency in flat spacetimes, the inertial spin connection is introduced as

$$\omega^a_{b\rho} := \Lambda^a_c \partial_\rho \Lambda_b^c,$$

where Λ^a_c denotes local Lorentz transformations, which include both rotations and boosts. Owing to the flatness of this spin connection, it is always possible to choose a local Lorentz frame in which the spin connection identically vanishes. Tetrads for which this condition holds are referred to as proper tetrads [17, 47], and in such a frame, the theory is formulated in the so-called Weitzenböck gauge.

The teleparallel connection (1) is not symmetric, with torsion given by,

$$T^\tau_{\rho\sigma} := \Gamma^\tau_{\sigma\rho} - \Gamma^\tau_{\rho\sigma}. \quad (4)$$

The contortion tensor $\mathcal{K}^{\rho\sigma}_\tau$, defined as the difference between the Levi-Civita and teleparallel connections, is expressed as,

$$\mathcal{K}^{\rho\sigma}_\tau := \frac{1}{2}(T^{\sigma\rho}_\tau + T_\tau^{\rho\sigma} - T^{\rho\sigma}_\tau). \quad (5)$$

Absolute parallelism requires $R^a_{bcd} = 0$, which holds in all frames and which is equivalent to the flatness condition (3). The Levi-Civita Ricci scalar $\bar{R}$ is related to the torsion scalar T by

$$e\bar{R} = -eT + 2\partial_\rho(eT_\sigma^{\sigma\rho}), \quad (6)$$

where $e = \det(e^a_\rho) = \sqrt{|g|}$ and T is defined as,

$$T = \frac{1}{4}T^{\rho\sigma\tau}T_{\rho\sigma\tau} + \frac{1}{2}T^{\rho\sigma\tau}T_{\tau\sigma\rho} - T_\sigma^{\sigma\rho}T_\tau^{\tau\rho}. \quad (7)$$

Thus, the Einstein-Hilbert action is equivalent to the Teleparallel Equivalent of General Relativity (TEGR) action up to a boundary term, as varying either with respect to their fundamental variables (metric vs. vierbeins) produces identical equations of motion [48].

Extending this to higher-order invariants, the teleparallel equivalent of the Gauss-Bonnet combination $\bar{G} = \bar{R}^2 - 4\bar{R}_{\rho\sigma}\bar{R}^{\rho\sigma} + \bar{R}_{\rho\sigma\tau\delta}\bar{R}^{\rho\sigma\tau\delta}$ is characterized by the torsion-based scalar T_G [20, 29],

$$e\bar{G} = eT_G - \partial_r \left[\mathcal{K}^{mn}{}_s \left(\mathcal{K}^{op}{}_{t,u} + \mathcal{K}^f{}_{tu} \mathcal{K}^{op}{}_f \right) \right] \delta_{mnop}^{rstu}. \quad (8)$$

The scalar T_G is constructed from the contortion tensor as

$$T_G = (\mathcal{K}^m{}_{er} \mathcal{K}^{en}{}_s \mathcal{K}^o{}_{ft} \mathcal{K}^{fp}{}_u - 2\mathcal{K}^{mn}{}_r \mathcal{K}^o{}_{es} \mathcal{K}^e{}_{ft} \mathcal{K}^{fp}{}_u + 2\mathcal{K}^{mn}{}_r \mathcal{K}^o{}_{es} \mathcal{K}^{ep}{}_f \mathcal{K}^f{}_{tu} + 2\mathcal{K}^{mn}{}_r \mathcal{K}^o{}_{es} \mathcal{K}^{ep}{}_{t,u}) \delta_{mnop}^{rstu}. \quad (9)$$

The term δ_{mnop}^{rstu} represents the generalized Kronecker delta, which in four dimensions admits the decomposition $\delta_{mnop}^{rstu} = \varepsilon^{rstu} \varepsilon_{mnop}$. This term is a functional of the contortion tensor and its first partial derivatives, exhibiting a quartic dependence on $\mathcal{K}$. This behavior is expected, as the curvature tensor is typically quadratic in contortion, and the Gauss-Bonnet invariant is itself a quadratic combination of curvature terms.

Varying the D-dimensional teleparallel Gauss-Bonnet action with respect to vierbeins yields equations identical to the metric variation of the standard Gauss-Bonnet action. Since T_G is a topological invariant in four dimensions, non-trivial dynamics are sought through the generalized action,

$$S_{TGB} = \int d^D x e \mathcal{F}(T, T_G). \quad (10)$$

The generalized Lagrangian density $\mathcal{F}(T, T_G)$ gives rise to non-trivial gravitational dynamics. In the special case $\mathcal{F}(T, T_G) = -T$, it reduces to TEGR, and thus to GR itself. Furthermore, for the choice $\mathcal{F}(T, T_G) = -T + c T_G$, where c denotes the Gauss-Bonnet coupling constant, the theory reproduces Einstein-Gauss-Bonnet gravity. The field equations obtained by performing a variation of the action (10) with respect to the tetrad fields are provided in Appendix A.

III. TETRADS FOR SPATIALLY CURVED COSMOLOGIES

To date, most research on teleparallel gravitational theories has focused on spatially flat FLRW cosmologies. This preference is driven not only by the observational relevance of flat geometries to the large-scale structure of the Universe but also by significant technical simplifications. In these contexts, the fundamental vierbein field $\{e^r\}$, related to the metric g via $g_{\rho\sigma} = \eta_{rs} e^r{}_\rho e^s{}_\sigma$, is often treated as the intuitive “square root” of the metric. Spatially curved FLRW Universes are usually described by the line element in spherical polar coordinates

$$ds^2 = -N^2 dt^2 + a^2 \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right], \quad (11)$$

where $N \equiv N(t)$ is the lapse function, $a \equiv a(t)$ is the scale factor governing the expansion of the universe, and k is the curvature parameter having values $k = 0, 1, -1$ representing flat, closed, and open cases respectively. For a spatially flat FLRW spacetime ($k = 0$) described by the line element (11) transformed in Cartesian coordinates

$$ds^2 = -N^2 dt^2 + a^2 (dx^2 + dy^2 + dz^2), \quad (12)$$

one typically adopts the diagonal tetrad

$$e^0{}_\rho dx^\rho = N dt, \quad e^1{}_\rho dx^\rho = a dx, \quad e^2{}_\rho dx^\rho = a dy, \quad e^3{}_\rho dx^\rho = a dz. \quad (13)$$

This choice is particularly effective because it ensures that the field equations derived from the $\mathcal{F}(T, T_G)$ action yield a consistent set of dynamical equations for the scale factor a . Under this tetrad, the torsion scalar T and the teleparallel Gauss-Bonnet equivalent T_G characterize the teleparallel Lagrangian. For the diagonal tetrad (13), the torsion scalar and teleparallel Gauss-Bonnet term are given by

$$T = \frac{6\dot{a}^2}{a^2 N^2} \stackrel{N \rightarrow 1}{=} 6H^2, \quad (14)$$

$$T_G = \frac{24\dot{a}^2 (N\ddot{a} - \dot{a}\dot{N})}{a^3 N^5} \stackrel{N \rightarrow 1}{=} 24H^2 (\dot{H} + H^2), \quad (15)$$

where $H = \dot{a}/a$ is the Hubble parameter and the lapse function is set to unity, i.e., $N = 1$. Furthermore, the field equations for the $\mathcal{F}(T, T_G)$ theory can be consistently derived from a reduced minisuperspace action by substituting these specific forms of the invariants into the general action.

Geometrically, the diagonal vierbein (13) reflects the autoparallel trajectories of flat Euclidean space. These trajectories are generated by the coordinate basis ∂_i , whose dual basis is simply dx^i . Thus, the frames describing these autoparallel lines are dx^i modulo a time-dependent scale factor. However, the implementation of $\mathcal{F}(T, T_G)$ gravity becomes considerably more complex in the context of closed and open FLRW universes. In Ref. [29], the authors obtained a good tetrad, but the teleparallel Gauss-Bonnet term obtained for open and closed FLRW line elements differ significantly and cannot yield a common expression in terms of curvature parameter k . We derive the explicit expressions for T and T_G in the presence of a nonvanishing spatial curvature parameter k , such that the resulting formulae are valid for open, closed and spatially flat cosmological geometries. Transforming (11) into hyperspherical coordinates using the transformation

$$r \mapsto \frac{\sin(\sqrt{k} R)}{\sqrt{k}} = \begin{cases} \sin R & \text{if } k = 1 \\ R & \text{if } k \rightarrow 0 \\ \sinh R & \text{if } k = -1 \end{cases}, \quad (16)$$

the line element for these geometries is expressed as,

$$ds^2 = -N^2 dt^2 + a^2 \left[dR^2 + \frac{\sin^2(\sqrt{k} R)}{k} (d\theta^2 + \sin^2 \theta d\phi^2) \right]. \quad (17)$$

For convenience, we define the parameter $\kappa = \sqrt{k}$, transforming (17) into

$$ds^2 = -N^2 dt^2 + \kappa^{-2} a^2 [d(\kappa R)^2 + \sin^2(\kappa R) (d\theta^2 + \sin^2 \theta d\phi^2)], \quad (18)$$

where $\kappa = 1$ for the closed universe, $\kappa = i$ for the open universe and the limit $\kappa \mapsto 0$ gives flat FLRW line element. Authors in [16] used the property $\kappa^4 = 1$ ($\kappa = 1, i$) to write $\kappa^{-2} = \kappa^2$ in (18), but this restricts the reduction to flat case. The naive square-root of metric method suggests the tetrad form

$$\begin{aligned} e^{0'}{}_\rho dx^\rho &= N dt, \\ e^{1'}{}_\rho dx^\rho &= a dR, \\ e^{2'}{}_\rho dx^\rho &= \kappa^{-1} a \sin(\kappa R) d\theta, \\ e^{3'}{}_\rho dx^\rho &= \kappa^{-1} a \sin(\kappa R) \sin \theta d\phi. \end{aligned} \quad (19)$$

However, this tetrad is considered “bad” [17], since this choice leads to a torsion scalar that depends explicitly on R ,

$$T = \frac{6\dot{a}^2}{a^2 N^2} - \frac{2\kappa^2 \cot^2(\kappa R)}{a^2}, \quad (20)$$

thereby breaking the homogeneity and isotropy of the background spacetime. In what follows, we obtain good tetrads for generic cosmological ansatz (11) by applying a local rotation $\mathfrak{R}$ to (19) parameterized by the Euler angles κR , $\pi - \theta$, and ϕ , with angular ranges $0 \leq \phi \leq 2\pi$, $0 \leq \theta \leq \pi$, and $0 \leq R \leq \pi$, where

$$[\mathfrak{R}] = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \phi & \sin \phi \\ 0 & 0 & -\sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\cos(\theta) & \sin(\theta) & 0 \\ 0 & -\sin(\theta) & -\cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos(\kappa R) & \sin(\kappa R) \\ 0 & 0 & -\sin(\kappa R) & \cos(\kappa R) \end{pmatrix},$$

thus relating the good tetrad e^a with naive choice (19) as

$$e^r = \mathfrak{R}^r_{s'} e^{s'}. \quad (21)$$

Geometrically, this rotation aligns the local frame with the natural parallel transport on the 3-sphere (or hyperbolic space) at each point. The naive tetrad (19) carries coordinate effects inherited from hyperspherical coordinates, while the Euler-angle rotation parameterized by $(\kappa R, \pi - \theta, \phi)$ compensates these inertial effects, ensuring that the resulting torsion invariants encode only the physical curvature of the background. The components of the resulting tetrads are given by

$$\begin{aligned} e^0{}_\rho dx^\rho &= N dt \\ e^1{}_\rho dx^\rho &= -a \cos \theta dR + \frac{1}{\kappa} a \sin \theta \sin(\kappa R) \cos(\kappa R) d\theta + \frac{1}{\kappa} a \sin^2 \theta \sin^2(\kappa R) d\phi \\ e^2{}_\rho dx^\rho &= -a \sin \theta \cos \phi dR \\ &\quad - \frac{1}{\kappa} a \sin(\kappa R) (\cos \theta \cos \phi \cos(\kappa R) + \sin \phi \sin(\kappa R)) d\theta \\ &\quad - \frac{1}{2\kappa} a \sin \theta (2 \cos \theta \cos \phi \sin^2(\kappa R) - \sin \phi \sin(2\kappa R)) d\phi \\ e^3{}_\rho dx^\rho &= a \sin \theta \sin \phi dR \\ &\quad + \frac{1}{\kappa} a \sin(\kappa R) (\cos \theta \sin \phi \cos(\kappa R) - \cos \phi \sin(\kappa R)) d\theta \\ &\quad + \frac{1}{\kappa} a \sin \theta \sin(\kappa R) (\cos \theta \sin \phi \sin(\kappa R) + \cos \phi \cos(\kappa R)) d\phi, \end{aligned} \quad (22)$$

which reproduces the metric given by (18). Note that the above tetrad remains non-singular, including the limit $\kappa \mapsto 0$. This choice, therefore, can be used to obtain quantities associated with the teleparallel connection [49]. The invariants (7) and (9) associated with (22) are

$$T = \frac{6\dot{a}^2}{a^2 N^2} - \frac{6\kappa^2}{a^2} \stackrel{N \rightarrow 1}{=} 6 \left(H^2 - \frac{\kappa^2}{a^2} \right), \quad (23)$$

$$T_G = \frac{24 (\dot{a}\dot{N} - N\ddot{a}) (\kappa^2 N^2 - \dot{a}^2)}{a^3 N^5} \stackrel{N \rightarrow 1}{=} 24 \left(H^2 + \dot{H} \right) \left(H^2 - \frac{\kappa^2}{a^2} \right). \quad (24)$$

Whereas the boundary terms on the RHS of (6) and (8) are given by

$$B := \frac{2}{e} \partial_\rho (e T_\sigma{}^{\sigma\rho}) = \frac{6\ddot{a}}{aN^2} + \frac{12\dot{a}^2}{a^2 N^2} - \frac{6\dot{a}\dot{N}}{aN^3} \stackrel{N \rightarrow 1}{=} 6 \left(3H^2 + \dot{H} \right), \quad (25)$$

$$B_G := \frac{1}{e} \partial_r \left[\mathcal{K}^{mn}{}_s \left(\mathcal{K}^{op}{}_{t,u} + \mathcal{K}^f{}_{tu} \mathcal{K}^{op}{}_f \right) \right] \delta^{rstu}_{mnop} = \frac{48\kappa^2 (\dot{a}\dot{N} - N\ddot{a})}{a^3 N^3} \stackrel{N \rightarrow 1}{=} -\frac{48\kappa^2 (H^2 + \dot{H})}{a^2}. \quad (26)$$

At this stage, we emphasize that the generic tetrads introduced in this work can be employed as suitable tetrads for a broader class of modified teleparallel gravity theories formulated on curved FLRW backgrounds. Given the large number of such models proposed in the literature, a systematic analysis of each of them lies beyond the scope of the present work. To substantiate the assertion that the tetrad (22) is applicable beyond the specific scenario considered in the main text, in Appendix B we derive the irreducible components of the torsion tensor (4). These components are employed in more fundamental and more general extensions of teleparallel gravity, such as New General Relativity [50] and teleparallel Horndeski gravity [51]. Additionally, we derive the Ricci scalar and the Gauss–Bonnet invariant associated with the metric given in equation (18),

$$\bar{R} = \frac{6\kappa^2}{a^2} - \frac{6\dot{a}\dot{N}}{aN^3} + \frac{6\dot{a}^2}{a^2N^2} + \frac{6\ddot{a}}{aN^2}, \quad (27)$$

$$\bar{G} = \frac{24(N\ddot{a} - \dot{a}\dot{N})(\kappa^2N^2 + \dot{a}^2)}{a^3N^5}. \quad (28)$$

The invariants derived above manifestly satisfy the equivalence relations $\bar{R} = -T + B$ and $\bar{G} = T_G - B_G$. It should be noted that differing sign conventions for the Levi-Civita symbol can lead some authors to write the latter condition as $\bar{G} = -T_G + B_G$ (see, for instance, [29]). Within the convention adopted here, one obtains $\bar{G} = T_G$ for a spatially flat FLRW ansatz. Consequently, theories of the form $-c_1T + f(T_G)$ and $c_1\bar{R} + f(\bar{G})$ are dynamically equivalent at the background level in such spacetimes, where c_1 is a constant. Nevertheless, as will be discussed below, this equivalence breaks down for FLRW cosmologies with non-vanishing spatial curvature and, more strongly, at the level of perturbations around any of these backgrounds [40, 44, 45]. As in the previous analysis, the lapse function is left arbitrary, whenever a specific choice for the lapse is taken, it will be stated explicitly.

It is straightforward to deduce from the definition $\kappa = \sqrt{k}$ that the expressions for the torsion scalar and teleparallel Gauss-Bonnet term are

$$T = \frac{6\dot{a}^2}{a^2N^2} - \frac{6k}{a^2}, \quad (29)$$

$$T_G = \frac{24(N\ddot{a} - \dot{a}\dot{N})(\dot{a}^2 - kN^2)}{a^3N^5}, \quad (30)$$

whereas the expression of the standard Gauss-Bonnet term G reduces to

$$\bar{G} = \frac{24(N\ddot{a} - \dot{a}\dot{N})(\dot{a}^2 + kN^2)}{a^3N^5}. \quad (31)$$

Consequently, actions that depend on arbitrary functions of these invariants will, in general, lead to distinct dynamical behaviors. As is evident from (30), any non-linear function of T_G involves second-order time derivatives of the scale factor a and therefore, generally the corresponding equations of motion for a theory with a non-linear $\mathcal{F}(T, T_G)$ are fourth order in a . Before proceeding to the analysis of the dynamics, we conclude this section by demonstrating that the tetrad (22) indeed constitutes a good tetrad for $\mathcal{F}(T, T_G)$ gravity. The components of the field equations (cf. A1) without any matter source are given by $\mathbb{E}_{\gamma\sigma} = 0$, where $\gamma, \sigma \in \{0, \hat{i} = 1, 2, 3\}$, as follows

$$\mathbb{E}_{00} \equiv -\frac{24Hk\dot{\mathcal{F}}_{T_G}}{a^2} + 24H^3\dot{\mathcal{F}}_{T_G} - T_G\mathcal{F}_{T_G} - 12H^2\mathcal{F}_T + \mathcal{F}, \quad (32)$$

$$\mathbb{E}_{\hat{i}\hat{j}} \equiv \left[8\left(H^2 - \frac{k}{a^2}\right)\ddot{\mathcal{F}}_{T_G} + 4\left(\frac{k}{a^2} - 3H^2 - \dot{H}\right)\mathcal{F}_T + 16H^3\dot{\mathcal{F}}_{T_G} + 16\dot{H}H\dot{\mathcal{F}}_{T_G} - T_G\mathcal{F}_{T_G} - 4H\dot{\mathcal{F}}_T + \mathcal{F} \right] \delta_{\hat{i}\hat{j}}, \quad (33)$$

where $T_G = 24\left(H^2 + \dot{H}\right)\left(H^2 - \frac{k}{a^2}\right)$ and the Lapse function is set to unity. We see that the non-diagonal parts of the field equations vanish, proving the “goodness” of the tetrad (22) a priori. Note that in the Weitzenböck gauge, the variation with respect to spin connection does not yield independent equations but conforms to the antisymmetric part of the field equations, which is satisfied here by construction. As previously discussed, the field equations may be of higher order. Consequently, it is advantageous to employ the Lagrange multiplier formalism in order to obtain an equivalent description of the theory in terms of an effective scalar field.

IV. MINISUPERSPACE DESCRIPTION OF THE ACTION

We focus on the model class of the form $-T + F(T_G)$. It constitutes the minimal framework in which the equivalence with $R + F(\bar{G})$ breaks down in non-flat spacetimes, as follows from Eqs. (30)–(31). In particular, torsion- and curvature-based Gauss-Bonnet invariants differ by curvature-dependent boundary contributions whenever $k \neq 0$. A comprehensive treatment of the general $\mathcal{F}(T, T_G)$ theory with arbitrary dependence on T is left for future investigation. By introducing a Lagrange multiplier, the nonlinear $F(T_G)$ contribution can be reformulated in terms of an equivalent

scalar-field representation, wherein the additional geometrodynamical degrees of freedom are encoded in an effective scalar mode. This reformulation provides a more manageable setting for performing stability analyses. Thus the action

$$S = \int d^4x e \left[-T + F(T_G) \right] \quad (34)$$

using the Lagrange multiplier ℓ , can be written as

$$S = \int d^4x e \left[-T + F(T_G) - \ell(T_G - \mathbb{G}) \right] \quad (35)$$

whose equations of motion impose the constraint $\mathbb{G} = T_G$, required for the analysis that follows. Varying the action with respect to T_G yields

$$\ell = F_{T_G} \quad (36)$$

transforming (35) into

$$S = \int d^4x e \left[-T + F(T_G) - T_G F_{T_G} + \mathbb{G} F_{T_G} \right]. \quad (37)$$

By treating $F(T_G)$ and its derivative F_{T_G} as independent quantities in the standard way, we introduce a scalar-field representation through

$$F_{T_G} = \phi, \quad (38)$$

$$\frac{F(T_G) - T_G F_{T_G}}{6} = V(\phi), \quad (39)$$

and upon imposing the constraint $\mathbb{G} = T_G$, obtain the action

$$S = \int d^4x e \left[-T + \phi T_G + 6V(\phi) \right]. \quad (40)$$

By substituting the torsion scalar and the teleparallel Gauss-Bonnet equivalent into (40) and subsequently applying integration by parts, we derive the following point-like Lagrangian

$$\mathcal{L}[N, a, \dot{a}, \phi, \dot{\phi}] = 6a^3 N V(\phi) + k \left(\frac{24\dot{a}\dot{\phi}}{N} + 6aN \right) - \frac{8\dot{a}^3 \dot{\phi}}{N^3} - \frac{6\dot{a}^2 a}{N}. \quad (41)$$

Here we have omitted the factor $\frac{\sin \theta \sin^2(\kappa R)}{\kappa^2}$, which appears in the tetrad determinant

$$e = \frac{a^3 N \sin \theta \sin^2(\kappa R)}{\kappa^2}, \quad (42)$$

since it does not influence the resulting field equations. By taking the variational derivative of the Lagrangian (41) with respect to the variables N , a , and ϕ , we obtain the corresponding evolution equations. It is evident that the lapse function N is non-dynamical, after performing integration by parts, no derivatives of N remain in the Lagrangian. Consequently, the associated constraint equation is given by

$$\frac{\partial \mathcal{L}}{\partial N} = 0 : \quad k \left(\frac{1}{a^2} - \frac{4H\dot{\phi}}{a^2} \right) + 4H^3 \dot{\phi} + H^2 + V(\phi) = 0 \quad (43)$$

The equations after variation with respect to a and ϕ respectively, are given by

$$\ddot{\phi} \left(2H^2 - \frac{2k}{a^2} \right) + \dot{\phi} \left(\frac{6Hk}{a^2} - 2H^3 + 4\dot{H}H \right) - \frac{k}{a^2} + \dot{H} = 0, \quad (44)$$

$$H^4 + \dot{H}H^2 - \frac{(H^2 + \dot{H})k}{a^2} + \frac{V_\phi}{4} = 0, \quad (45)$$

where Lapse function is set to unity and $V_\phi = \frac{dV}{d\phi}$. We now proceed with the analysis of the dynamical behavior of the system of equations (43)-(45).

V. THE DYNAMICS

In the analysis of cosmological models, it is standard practice to introduce auxiliary variables that recast the governing equations into an autonomous dynamical system. This procedure leads to a system of the form $\mathbf{X}' = \mathbf{F}(\mathbf{X})$, where $\mathbf{X}$ denotes the column vector of auxiliary variables and $\mathbf{F}(\mathbf{X})$ represents the corresponding autonomous vector field. Here, the prime indicates differentiation with respect to a reparameterized time to render the derived system autonomous. In this framework, we introduce the following dimensionless phase-space variables

$$\mathcal{X} = \frac{4\dot{\phi}}{\sqrt{1+H^2}}, \quad (46a)$$

$$\mathcal{Y} = \frac{V(\phi)}{1+H^2}, \quad (46b)$$

$$\mathcal{Z} = \frac{H}{\sqrt{1+H^2}}, \quad (46c)$$

$$\Omega_k = \frac{k}{a^2(1+H^2)}, \quad (46d)$$

$$\lambda = \frac{V_\phi}{V}. \quad (46e)$$

This methodology departs from the conventional H -normalization framework by introducing a parametrization that remains well-defined and stable at the transition point $H = 0$. In this way, the formulation guarantees a non-singular dynamical evolution through the epoch of vanishing expansion rate, thereby permitting a mathematically consistent change in the sign of H . In terms of these variables, the constraint equation (43) becomes

$$\frac{\mathcal{X}\mathcal{Z}(\mathcal{Z}^2 - \Omega_k)}{1 - \mathcal{Z}^2} + \Omega_k + \mathcal{Y} + \mathcal{Z}^2 = 0. \quad (47)$$

We now establish the dynamical equations for each constituent variable. By introducing the reformulated evolution parameter $d\Sigma = \sqrt{1+H^2}dt$, the governing equations for the set $\{\mathcal{X}, \mathcal{Y}, \mathcal{Z}, \Omega_k, \lambda\}$ are cast as a closed system of first-order autonomous differential equations,

$$\begin{aligned} \frac{d\mathcal{X}}{d\Sigma} = & 1 - \mathcal{Z}^2 + \mathcal{X}\mathcal{Z}^3 + \frac{\lambda\mathcal{X}\mathcal{Y}\mathcal{Z}^3 - \lambda\mathcal{X}\mathcal{Y}\mathcal{Z} - 12\mathcal{Y}\mathcal{Z}^2 + 12\mathcal{Y} - 8\mathcal{Z}^4 + 8\mathcal{Z}^2}{4(\Omega_k - \mathcal{Z}^2)} \\ & + \frac{-\lambda\mathcal{X}\mathcal{Y}\mathcal{Z}^3 + \lambda\mathcal{X}\mathcal{Y}\mathcal{Z} + \lambda\mathcal{Y} + \lambda\mathcal{Y}\mathcal{Z}^4 - 2\lambda\mathcal{Y}\mathcal{Z}^2}{2(\Omega_k - \mathcal{Z}^2)^2}, \end{aligned} \quad (48a)$$

$$\frac{d\mathcal{Y}}{d\Sigma} = \frac{1}{4}\mathcal{Y} \left(\frac{-2\mathcal{Z}^3(4\Omega_k + \lambda\mathcal{Y}) + 2\lambda\mathcal{Y}\mathcal{Z} + 8\mathcal{Z}^5}{\mathcal{Z}^2 - \Omega_k} + \lambda\mathcal{X} \right), \quad (48b)$$

$$\frac{d\mathcal{Z}}{d\Sigma} = \frac{1}{4}(\mathcal{Z} - 1)(\mathcal{Z} + 1) \left(\frac{\lambda\mathcal{Y}(\mathcal{Z} - 1)(\mathcal{Z} + 1)}{\Omega_k - \mathcal{Z}^2} + 4\mathcal{Z}^2 \right), \quad (48c)$$

$$\frac{d\Omega_k}{d\Sigma} = \frac{\mathcal{Z}(\mathcal{Z}^2 - 1)\Omega_k(-4\Omega_k - \lambda\mathcal{Y} + 4\mathcal{Z}^2)}{2(\mathcal{Z}^2 - \Omega_k)}, \quad (48d)$$

$$\frac{d\lambda}{d\Sigma} = \frac{\lambda^2}{4}\mathcal{X}\Lambda(\lambda), \quad (48e)$$

where $\Lambda = \frac{VV_{\phi\phi}}{V_\phi^2} - 1$. The scalar-field potential must be chosen such that Λ remains a single-valued function. By construction, $\mathcal{Z}$ is constrained within the range $\mathcal{Z}^2 \leq 1$, although the remaining variables are not compactified. Specifically, the case $\mathcal{Z}^2 = 1$ characterizes a scaling universe, whereas $\mathcal{Z}^2 < 1$ corresponds to a de Sitter solution. The sign of $\mathcal{Z}$ dictates the global dynamics: for $\mathcal{Z} > 0$, the system describes an expanding universe ($H > 0$), while for $\mathcal{Z} < 0$, it describes a collapsing universe ($H < 0$). The deceleration parameter can be formulated in terms of the underlying dynamical variables as follows

$$q = -1 - \frac{\dot{H}}{H^2} \equiv \frac{\lambda\mathcal{Y}(1 - \mathcal{Z}^2)}{4\mathcal{Z}^2(\mathcal{Z}^2 - \Omega_k)}. \quad (49)$$

We now comment on the system of differential equations (48). First, the dynamical variables are subject to the constraint equation (47), therefore reducing the dimension of the dynamical system to four. Without loss of generality,

we eliminate $\mathcal{Y}$ from the system using the constraint equation. Second, as previously discussed, the choice of the scalar field potential is crucial for obtaining the system in closed form. For instance, in certain inflationary scenarios featuring potentials such as $V(\phi) \propto \phi^a \ln^b(\phi)$ [52] or $V(\phi) \propto \phi^m e^{-p\phi^n}$ [53], the parameter Λ cannot be represented as a single-valued function of λ . Choosing an exponential potential given by $V(\phi) = V_0 e^{\lambda_0 \phi}$ results in $\Lambda(\lambda) = 0$ and thus $\lambda = \lambda_0$, further reducing the dimension of the system to three. For the exponential potential, the singular solution of Clairaut's equation obtained from (38)-(39) is given by

$$F(T_G) = \frac{T_G}{\lambda_0} \left[\log_e \left(-\frac{T_G}{6\lambda_0 V_0} \right) - 1 \right]. \quad (50)$$

For the remainder of the analysis, the subscript 0 is omitted from λ_0 . Consequently, λ is to be treated as a non-zero constant throughout the subsequent derivations. With the above assumptions, the dynamical system reduces to

$$\begin{aligned} \frac{d\mathcal{X}}{d\Sigma} = & -\frac{\lambda(\mathcal{Z}-1)(\mathcal{Z}+1)\mathcal{Z}^2(-\mathcal{X}\mathcal{Z}+\mathcal{Z}^2-1)}{(\Omega_k-\mathcal{Z}^2)^2} + \frac{-\lambda-\lambda\mathcal{X}\mathcal{Z}^5+\lambda\mathcal{X}\mathcal{Z}^3+\lambda\mathcal{X}^2\mathcal{Z}^2-\lambda\mathcal{Z}^4+2\lambda\mathcal{Z}^2+8\mathcal{Z}^4-8\mathcal{Z}^2}{2(\Omega_k-\mathcal{Z}^2)} \\ & + \frac{1}{4}(-\lambda\mathcal{X}\mathcal{Z}^3-\lambda\mathcal{X}^2\mathcal{Z}^2+\lambda\mathcal{X}\mathcal{Z}+4\mathcal{X}\mathcal{Z}^3+12\mathcal{X}\mathcal{Z}+8\mathcal{Z}^2-8), \end{aligned} \quad (51a)$$

$$\frac{d\mathcal{Z}}{d\Sigma} = \frac{-\lambda\mathcal{Z}^6+2\lambda\mathcal{Z}^4-\lambda\mathcal{Z}^2}{2(\Omega_k-\mathcal{Z}^2)} - \frac{1}{4}(\mathcal{Z}^2-1)(-\lambda+\lambda\mathcal{X}\mathcal{Z}+\lambda\mathcal{Z}^2-4\mathcal{Z}^2), \quad (51b)$$

$$\frac{d\Omega_k}{d\Sigma} = -\frac{1}{2}\mathcal{Z}\Omega_k(-\lambda+\lambda\mathcal{X}\mathcal{Z}+\lambda\mathcal{Z}^2-4\mathcal{Z}^2+4) + \frac{\lambda\mathcal{Z}^5-\lambda\mathcal{Z}^7}{\Omega_k-\mathcal{Z}^2} - \lambda(\mathcal{Z}^2-1)\mathcal{Z}^3. \quad (51c)$$

The system under consideration is strongly nonlinear and therefore does not admit a closed-form analytical solution. To analyze its local behavior in the vicinity of an equilibrium, we consider the above autonomous dynamical system in the form

$$\mathbf{X}' = \mathbf{F}(\mathbf{X}), \quad (52)$$

where $\mathbf{X} = \{\mathcal{X}, \mathcal{Z}, \Omega_k\}$ and $\mathbf{F} \in C^r(U_c)$ with $r \geq 1$, defined on a neighborhood U_c of a critical point $\mathbf{X}_c$. Since $\mathbf{F}$ is sufficiently differentiable on U_c , it admits a Taylor expansion about $\mathbf{X}_c$ of the form

$$\mathbf{F}(\mathbf{X}) = \mathbf{F}(\mathbf{X}_c) + \mathcal{J}(\mathbf{X}_c)(\mathbf{X} - \mathbf{X}_c) + \mathcal{O}(|\mathbf{X} - \mathbf{X}_c|^2), \quad (53)$$

where $\mathcal{J}(\mathbf{X}_c) = D\mathbf{F}(\mathbf{X}_c)$ denotes the Jacobian (stability) matrix evaluated at the equilibrium point. Because $\mathbf{X}_c$ is a critical point, one has $\mathbf{F}(\mathbf{X}_c) = 0$, and thus, in a sufficiently small neighborhood of $\mathbf{X}_c$, the dynamics are, to leading order, governed by the linearized system

$$\mathbf{X}' \approx \mathcal{J}(\mathbf{X}_c)(\mathbf{X} - \mathbf{X}_c). \quad (54)$$

The Hartman–Grobman theorem [54, 55] rigorously justifies this linear approximation: if $\mathbf{X}_c$ is a hyperbolic equilibrium of the nonlinear system (52), then its local phase portrait is topologically conjugate to that of the linearized system (54). That is, there exists a homeomorphism $H : U_c \rightarrow V$, with V an open neighborhood of the origin, mapping trajectories of the nonlinear system onto those of the linear system in an orientation-preserving way. Hence, the qualitative stability near $\mathbf{X}_c$ is completely determined by the eigenstructure of $\mathcal{J}(\mathbf{X}_c)$. A stronger result by Hartman [56, 57] states that if $\mathbf{F} \in C^2(U_c)$ and $\mathbf{X}_c$ is hyperbolic, then this conjugacy can be taken as a C^1 -diffeomorphism, so both the topological and local differentiable structures of the flow are preserved near the critical point. We therefore analyze in detail the linearized system (51) for both flat and curved geometries, based on the critical points of the full nonlinear system.

A. Dynamics of Flat Universe

The flat case corresponds to $k = 0$, which in the language of dynamical variables, transforms to $\Omega_k = 0$. This further reduces the dimension by 1, yielding the following 2D–dynamical system

$$\frac{d\mathcal{X}}{d\Sigma} = \frac{\lambda(\mathcal{X}\mathcal{Z}^5 - \mathcal{X}^2\mathcal{Z}^4 + 3\mathcal{X}\mathcal{Z}^3 - 2\mathcal{X}^2\mathcal{Z}^2 - 4\mathcal{X}\mathcal{Z} - 2\mathcal{Z}^4 + 4\mathcal{Z}^2 - 2)}{4\mathcal{Z}^2} + \mathcal{X}\mathcal{Z}^3 + 3\mathcal{X}\mathcal{Z} - 2\mathcal{Z}^2 + 2, \quad (55a)$$

$$\frac{d\mathcal{Z}}{d\Sigma} = \frac{1}{4}(\mathcal{Z}-1)(\mathcal{Z}+1)(-\lambda-\lambda\mathcal{X}\mathcal{Z}+(\lambda+4)\mathcal{Z}^2). \quad (55b)$$

The critical points/ stationary points $(\mathcal{X}_c, \mathcal{Z}_c)$ are

$$\begin{aligned}\dot{\mathcal{P}}_1 &: (0, -1), & \dot{\mathcal{P}}_2 &: \left(-\frac{16}{3\lambda}, -1\right), \\ \dot{\mathcal{P}}_3 &: (0, 1), & \dot{\mathcal{P}}_4 &: \left(\frac{16}{3\lambda}, 1\right), \\ \dot{\mathcal{P}}_5 &: \left(0, -\sqrt{\frac{\lambda}{\lambda+4}}\right), & \dot{\mathcal{P}}_6 &: \left(0, \sqrt{\frac{\lambda}{\lambda+4}}\right).\end{aligned}$$

We now explain the behavior of each critical point in detail. With the eigenvalues of a critical point, we mean the eigenvalues of the corresponding Jacobian matrix.

- The critical point $\dot{\mathcal{P}}_1 = (0, -1)$ exists for all $\lambda \in \mathbb{R}$. Its eigenvalues, $(-4, -2)$, are both negative, implying that the point is a stable attractor. Since $\mathcal{Z}^2 = 1$, the solution is asymptotic and corresponds to a collapsing phase. At this point, $q(\dot{\mathcal{P}}_1) = 0$ and $\mathcal{Y}(\dot{\mathcal{P}}_1) = -1$.
- The point $\dot{\mathcal{P}}_2 = \left(-\frac{16}{3\lambda}, -1\right)$ exists for $\lambda \neq 0$. The eigenvalues $\left(\frac{2}{3}, 4\right)$ are positive, indicating that the point is an unstable repeller. The corresponding deceleration parameter is $q = -\frac{4}{3}$, characterizing an accelerated regime that may be associated with an early inflationary phase.
- The critical point $\dot{\mathcal{P}}_3 = (0, 1)$ exists for all $\lambda \in \mathbb{R}$. Its eigenvalues $(4, 2)$ are positive, showing that it is an unstable repeller. As in $\dot{\mathcal{P}}_1$, the condition $\mathcal{Z}^2 = 1$ indicates an asymptotic solution. However, here the evolution corresponds to an expanding phase. One obtains $q(\dot{\mathcal{P}}_3) = 0$ and $\mathcal{Y}(\dot{\mathcal{P}}_3) = -1$.
- The point $\dot{\mathcal{P}}_4 = \left(\frac{16}{3\lambda}, 1\right)$ is defined for $\lambda \neq 0$. Its eigenvalues $(-4, -\frac{2}{3})$ are negative, therefore the point is a stable attractor. The value $q = -\frac{4}{3}$ corresponds to a late-time accelerated solution, compatible with a phantom-dominated dark energy regime.
- The critical point $\dot{\mathcal{P}}_5 = \left(0, -\sqrt{\frac{\lambda}{\lambda+4}}\right)$ exists for $\lambda < -4$ or $\lambda \geq 0$. Its eigenvalues are

$$\left(-\frac{1}{2}(\sqrt{17}-3)\sqrt{\frac{\lambda}{\lambda+4}}, \frac{1}{2}(3+\sqrt{17})\sqrt{\frac{\lambda}{\lambda+4}}\right),$$

which have opposite signs, implying saddle behaviour. In this case the scalar-field potential contributes non-trivially, with $\mathcal{Y} \rightarrow -\frac{\lambda}{\lambda+4}$. The deceleration parameter takes the value $q = -1$, corresponding to a de Sitter like transient solution driven by an effective cosmological constant.

- The point $\dot{\mathcal{P}}_6 = \left(0, \sqrt{\frac{\lambda}{\lambda+4}}\right)$ exists under the same conditions, namely $\lambda < -4$ or $\lambda \geq 0$. Its eigenvalues are

$$\left(-\frac{1}{2}(3+\sqrt{17})\sqrt{\frac{\lambda}{\lambda+4}}, \frac{1}{2}(\sqrt{17}-3)\sqrt{\frac{\lambda}{\lambda+4}}\right),$$

and again have opposite signs, showing that the point is a saddle. As for $\dot{\mathcal{P}}_5$, one has $\mathcal{Y} \rightarrow -\frac{\lambda}{\lambda+4}$, while $q = -1$ identifies a de Sitter like transient phase. The condition $\mathcal{Z}^2 \leq 1$, imposes condition $\lambda \geq 0$.

In addition to the equilibrium points identified in the finite phase-space region, given by system (55), it is necessary to investigate the asymptotic structure of the dynamical system, since the variable $\mathcal{X}$ may diverge and thus allow for equilibrium configurations at infinity. To incorporate this sector into a compact phase space, we introduce the compactified variable $\bar{\mathcal{X}}$ through the transformation

$$\mathcal{X} = \frac{\bar{\mathcal{X}}}{\sqrt{1-\bar{\mathcal{X}}^2}}, \quad (56)$$

which maps the asymptotic boundary to $\bar{\mathcal{X}}^2 = 1$.

Within this compactified formulation, the system admits four equilibrium points at infinity, denoted by $\dot{\mathcal{P}}^\infty = (\bar{\mathcal{X}}(\dot{\mathcal{P}}^\infty), \mathcal{Z}(\dot{\mathcal{P}}^\infty))$, with coordinates

$$\dot{\mathcal{P}}_1^{\infty \pm} = (\pm 1, 1), \quad \dot{\mathcal{P}}_2^{\infty \pm} = (\pm 1, -1). \quad (57)$$

At these asymptotic critical points, the deceleration parameter becomes unbounded. Specifically, one obtains

$$q\left(\dot{\mathcal{P}}_1^{\infty\pm}\right) = -\text{sign}\left[\lambda\bar{\mathcal{X}}\left(\dot{\mathcal{P}}_1^{\infty\pm}\right)\right] \infty, \quad (58)$$

and

$$q\left(\dot{\mathcal{P}}_2^{\infty\pm}\right) = \text{sign}\left[\lambda\bar{\mathcal{X}}\left(\dot{\mathcal{P}}_2^{\infty\pm}\right)\right] \infty. \quad (59)$$

The points $\dot{\mathcal{P}}_1^{\infty\pm}$ correspond to expanding cosmological branches. However, whenever $\text{sign}\left[\lambda\bar{\mathcal{X}}\left(\dot{\mathcal{P}}_2^{\infty\pm}\right)\right] < 0$, the deceleration parameter becomes positive, indicating that these asymptotic configurations describe solutions approaching Minkowski spacetime in the asymptotic limit. A similar behavior is found for $\dot{\mathcal{P}}_2^{\infty\pm}$ whenever $\text{sign}\left[\lambda\bar{\mathcal{X}}\left(\dot{\mathcal{P}}_2^{\infty\pm}\right)\right] > 0$.

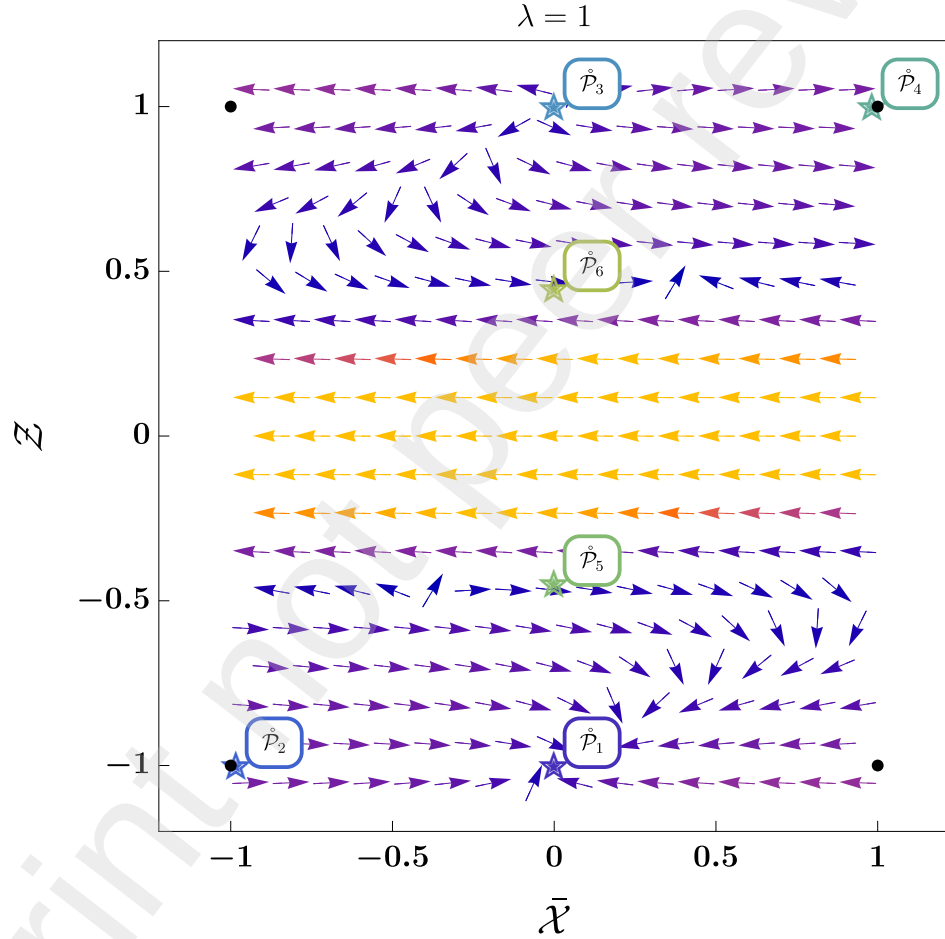


Figure 1. Phase portrait of the dynamical system (55) represented in the compactified variables $\bar{\mathcal{X}}$ and $\mathcal{Z}$. Points at infinity are denoted by black dots, while all critical points located in the finite region are explicitly depicted and labeled within boxes. It is observed that $\dot{\mathcal{P}}_1$ and $\dot{\mathcal{P}}_4$ are always attractors, $\dot{\mathcal{P}}_2$ and $\dot{\mathcal{P}}_3$ act as sources, and $\dot{\mathcal{P}}_5$ and $\dot{\mathcal{P}}_6$ correspond to saddle points.

The existence of equilibrium points at infinity is therefore essential for understanding transitions between expanding and contracting cosmological phases. Since $\mathcal{Z} = 0$ defines a singular hypersurface of the dynamical system, such transitions cannot occur through the finite region of phase space and must instead proceed through the asymptotic sector. This behavior is explicitly illustrated in Fig. 1. Finally, all equilibrium points located at infinity are found to be unstable, indicating that they act as repelling asymptotic states in the cosmological evolution. All of the above is summarized in Table I.

Table I. Critical points, existence conditions, eigenvalues, deceleration parameter, and dynamical classification.

Label	$(\mathcal{X}_c, \mathcal{Z}_c)$	Existence	Eigenvalues	q	Stability
$\dot{\mathcal{P}}_1$	$(0, -1)$	$\lambda \in \mathbb{R}$	$(-4, -2)$	0	Attractor
$\dot{\mathcal{P}}_2$	$(-\frac{16}{3\lambda}, -1)$	$\lambda \neq 0$	$(\frac{2}{3}, 4)$	$-\frac{4}{3}$	Repeller
$\dot{\mathcal{P}}_3$	$(0, 1)$	$\lambda \in \mathbb{R}$	$(4, 2)$	0	Repeller
$\dot{\mathcal{P}}_4$	$(\frac{16}{3\lambda}, 1)$	$\lambda \neq 0$	$(-4, -\frac{2}{3})$	$-\frac{4}{3}$	Attractor
$\dot{\mathcal{P}}_5$	$(0, -\sqrt{\frac{\lambda}{\lambda+4}})$	$\lambda < -4 \vee \lambda \geq 0$	$[-\frac{1}{2}(\sqrt{17}-3)\sqrt{\frac{\lambda}{\lambda+4}}, \frac{1}{2}(3+\sqrt{17})\sqrt{\frac{\lambda}{\lambda+4}}]$	-1	Saddle
$\dot{\mathcal{P}}_6$	$(0, \sqrt{\frac{\lambda}{\lambda+4}})$	$\lambda < -4 \vee \lambda \geq 0$	$[-\frac{1}{2}(3+\sqrt{17})\sqrt{\frac{\lambda}{\lambda+4}}, \frac{1}{2}(\sqrt{17}-3)\sqrt{\frac{\lambda}{\lambda+4}}]$	-1	Saddle
$\dot{\mathcal{P}}_1^{\infty\pm}$	$(\pm 1, 1)$	$\lambda \in \mathbb{R}$	-	$-\text{sign}[\lambda \mathcal{X}(\dot{\mathcal{P}}_1^{\infty\pm})] \infty$	Saddle
$\dot{\mathcal{P}}_2^{\infty\pm}$	$(\pm 1, -1)$	$\lambda \in \mathbb{R}$	-	$\text{sign}[\lambda \mathcal{X}(\dot{\mathcal{P}}_2^{\infty\pm})] \infty$	Saddle

B. Dynamics with spatial Curvature

We now consider the full three-dimensional system (51), in which the curvature variable Ω_k is allowed to evolve dynamically. The critical points for the system are given by

$$\begin{aligned}
 \dot{\mathcal{P}}_1^k &: (0, -1, \Omega_k), & \dot{\mathcal{P}}_2^k &: (0, 1, \Omega_k), \\
 \dot{\mathcal{P}}_3^k &: \left(-\frac{16}{3\lambda}, -1, 0\right), & \dot{\mathcal{P}}_4^k &: \left(\frac{16}{3\lambda}, 1, 0\right), \\
 \dot{\mathcal{P}}_5^k &: \left(0, -\sqrt{\frac{\lambda}{\lambda+4}}, 0\right), & \dot{\mathcal{P}}_6^k &: \left(0, \sqrt{\frac{\lambda}{\lambda+4}}, 0\right).
 \end{aligned}$$

We see that the system contains two continuous equilibrium lines $\dot{\mathcal{P}}_1^k$ and $\dot{\mathcal{P}}_2^k$ parametrized by Ω_k . Apart from these two points, all other critical points are confined to the $\mathcal{XZ}$ -plane. The stability properties of the remaining critical points can be compared directly with those of the flat system as they lie on the invariant submanifold $\Omega_k = 0$. Nevertheless, we analyze all the points below

- The critical point $\dot{\mathcal{P}}_1^k = (0, -1, \Omega_k)$ exists for all $\lambda \in \mathbb{R}$ and $\Omega_k \in \mathbb{R}$. Eigenvalues of the Jacobian at this point are $(-4, -2, 0)$, thus classifying it as a non-hyperbolic equilibrium point. At this point, we have $\mathcal{Z} < 0$, thus a collapsing asymptotic solution is represented, while the potential takes the value $\mathcal{V} = -1 - \Omega_k$. The zero eigenvalue encodes the arbitrariness in Ω_k , implying that the stability properties in that direction cannot be determined at the linear approximation. The deceleration parameter is $q = 0$ in this line of critical points.

To investigate the stability properties, we perform a regular perturbation analysis to obtain an exact solution for the evolution of small deviations from the critical configuration. We introduce time-dependent perturbations around the critical coordinates up to first order, defined as $\mathcal{X}(\Sigma) = 0 + \delta\mathcal{X}(\Sigma)$, $\mathcal{Z}(\Sigma) = -1 + \delta\mathcal{Z}(\Sigma)$, and $\Omega_k(\Sigma) = \Omega_{k0} + \delta\Omega_k(\Sigma)$, where Ω_{k0} denotes the background spatial curvature parameter along the continuous equilibrium line. Upon integrating the dynamical system, we obtain the following exact asymptotic behavior for the perturbations,

$$\delta\mathcal{Z} \simeq e^{-2\Sigma}, \quad (60a)$$

$$\delta\mathcal{X} \simeq \mathcal{O}(e^{-2\Sigma}) + \mathcal{O}(e^{-4\Sigma}), \quad (60b)$$

$$\delta\Omega_k \simeq \mathcal{O}(e^{-2\Sigma}) + \mathcal{O}(e^{-4\Sigma}) + C. \quad (60c)$$

The complete analytic expressions are provided in Appendix C. Since all time-dependent contributions decay with strictly negative exponents, none of the perturbations exhibit unbounded growth. This demonstrates that the equilibrium line is a stable, normally hyperbolic attractor in cosmological phase space.

- The critical point $\mathcal{P}_2^k = (0, 1, \Omega_k)$ also exists for all $\lambda \in \mathbb{R}$ and $\Omega_k \in \mathbb{R}$, and possesses eigenvalues $(2, 4, 0)$, which again characterize it as a non-hyperbolic point but since the other eigenvalues are positive it is not stable. The potential evaluates to $\mathcal{Y} = -1 - \Omega_k$, and, analogously to $\mathcal{P}_1^k$, the solution is asymptotic with $\mathcal{Z}^2 = 1$. However, the sign of $\mathcal{Z}$ now corresponds to an expanding configuration rather than a collapsing one. The deceleration parameter $q = 0$ indicates that this point corresponds to a past transitional phase, from which the positive eigenvalues repel trajectories.
- The point $\mathcal{P}_3^k = (-\frac{16}{3\lambda}, -1, 0)$, defined for $\lambda \neq 0$, acts as an unstable repeller, with eigenvalues $(4, \frac{2}{3}, \frac{8}{3})$. The vanishing curvature parameter $\Omega_k = 0$ reduces the dynamics to the spatially flat sub-manifold, and the value of the deceleration parameter $q = -\frac{4}{3}$ places this solution in an accelerated regime, typical of an early-time inflationary or pre-radiation-dominated source.
- The critical point $\mathcal{P}_4^k = (\frac{16}{3\lambda}, 1, 0)$, valid for $\lambda \neq 0$, is a stable attractor with eigenvalues $(-4, -\frac{2}{3}, -\frac{8}{3})$. Here again, the curvature contribution vanishes, and the corresponding deceleration parameter $q = -\frac{4}{3}$ indicates a strongly accelerated asymptotic expansion.
- The points $\mathcal{P}_5^k$ and $\mathcal{P}_6^k$ are saddle points with $q = -1$, corresponding to transient de Sitter-like configurations. Both lie on the flat submanifold $\Omega_k = 0$.

All of the above observations are summarized in Table II. Employing the same compactified variable $\bar{\mathcal{X}}$, we determine the critical points at infinity as

$$\mathcal{P}_1^{k\infty\pm} = (\pm 1, 1, \Omega_k), \quad \mathcal{P}_2^{k\infty\pm} = (\pm 1, -1, \Omega_k). \quad (61)$$

Their qualitative interpretation is unchanged from the flat system, the asymptotic points remain unstable. The three-dimensional phase portrait in Fig. 2 is displayed in uncompactified $\mathcal{X}$ for visual clarity, the compactified portrait is qualitatively identical in the finite region and confirms that all points at infinity are unstable, in agreement with the flat-case analysis.

We find that most of the finite critical points associated with accelerated asymptotic regimes are confined to the $\Omega_k = 0$ hyperplane. The points $\mathcal{P}_5^k$ and $\mathcal{P}_6^k$ correspond to states dominated by the scalar-field potential energy, characteristic of inflationary-like evolution. Since these points lie on the flat submanifold, trajectories that approach them can experience a transient suppression of the curvature contribution. In contrast, trajectories within the basin of $\mathcal{P}_4^k$ asymptotically approach the flat submanifold, with $\Omega_k \rightarrow 0$. The curved attractor $\mathcal{P}_1^k$, however, admits asymptotic states with nonvanishing spatial curvature. Moreover, the sign of the initial curvature contribution is preserved along trajectories approaching this attractor. Overall, the dynamical analysis demonstrates that incorporating spatial curvature into the teleparallel Gauss-Bonnet framework yields a substantially richer phase-space structure than in the spatially flat configuration.

Table II. Critical points, existence conditions, eigenvalues, deceleration parameter, and dynamical classification.

Label	$(\mathcal{X}_c, \mathcal{Z}_c, \Omega_{k_c})$	Existence	Eigenvalues	q	Stability
$\mathcal{P}_1^k$	$(0, -1, \Omega_k)$	$\lambda \in \mathbb{R} \wedge \Omega_k \in \mathbb{R}$	$(-4, -2, 0)$	0	Attractor
$\mathcal{P}_2^k$	$(0, 1, \Omega_k)$	$\lambda \in \mathbb{R} \wedge \Omega_k \in \mathbb{R}$	$(4, 2, 0)$	0	Unstable
$\mathcal{P}_3^k$	$(-\frac{16}{3\lambda}, -1, 0)$	$\lambda \neq 0$	$(4, \frac{2}{3}, \frac{8}{3})$	$-\frac{4}{3}$	Repeller
$\mathcal{P}_4^k$	$(\frac{16}{3\lambda}, 1, 0)$	$\lambda \neq 0$	$(-4, -\frac{2}{3}, -\frac{8}{3})$	$-\frac{4}{3}$	Attractor
$\mathcal{P}_5^k$	$(0, -\sqrt{\frac{\lambda}{\lambda+4}}, 0)$	$\lambda < -4 \vee \lambda \geq 0$	$[-\frac{1}{2}(\sqrt{17}-3)\sqrt{\frac{\lambda}{\lambda+4}}, \frac{1}{2}(3+\sqrt{17})\sqrt{\frac{\lambda}{\lambda+4}}, 2\sqrt{\frac{\lambda}{\lambda+4}}]$	-1	Saddle
$\mathcal{P}_6^k$	$(0, \sqrt{\frac{\lambda}{\lambda+4}}, 0)$	$\lambda < -4 \vee \lambda \geq 0$	$[-\frac{1}{2}(3+\sqrt{17})\sqrt{\frac{\lambda}{\lambda+4}}, \frac{1}{2}(\sqrt{17}-3)\sqrt{\frac{\lambda}{\lambda+4}}, -2\sqrt{\frac{\lambda}{\lambda+4}}]$	-1	Saddle

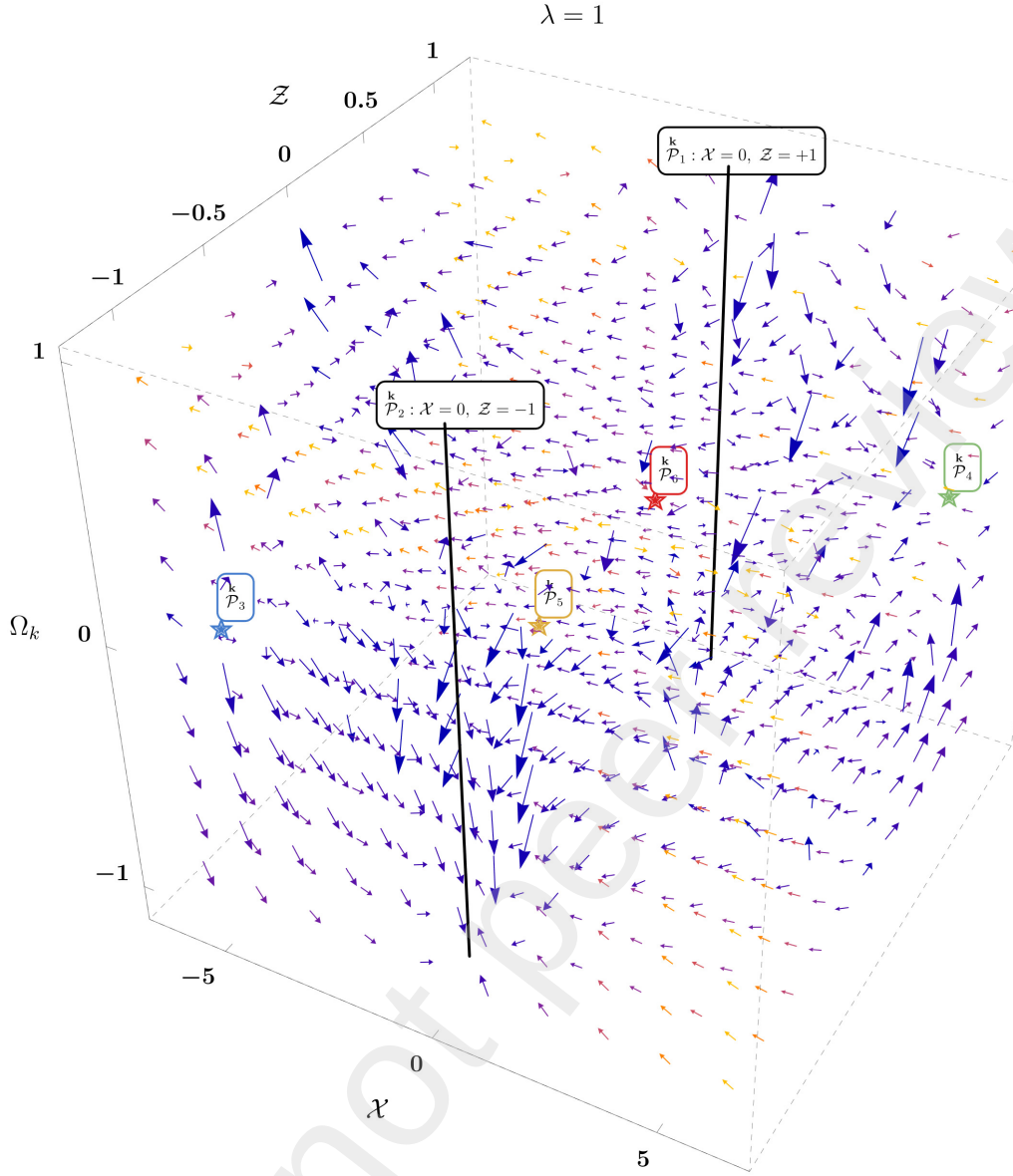


Figure 2. Three-dimensional phase-space portrait of the dynamical system (51) in the variables $\mathcal{X}$, $\mathcal{Z}$, and Ω_k . Critical points in the invariant plane $\Omega_k = 0$ are shown with boxed labels, while the critical lines $\mathcal{P}_1$ and $\mathcal{P}_2$ are labeled at their endpoints. The line $\mathcal{P}_1$ is a normally hyperbolic attractor, whereas $\mathcal{P}_2$ is an unstable manifold repelling nearby trajectories. The point $\mathcal{P}_3$ is a source, and $\mathcal{P}_4$ is an attractor. The points $\mathcal{P}_5$ and $\mathcal{P}_6$ are saddle points.

VI. CONCLUSION

The central technical contribution of this work is the construction of tetrads (22) for FLRW spacetimes that explicitly incorporate the spatial curvature parameter k within a single unified expression, simultaneously valid for open $k = -1$, flat $k = 0$, and closed $k = +1$ geometries. The construction is obtained through a suitable local Lorentz transformation of the diagonal tetrad and preserves the homogeneity and isotropy of the torsion invariants. The admissibility of these tetrads is established by the vanishing of the antisymmetric sector of the field equations. Their regular behavior in the flat limit $\kappa \rightarrow 0$ provides a smooth and consistent recovery of the standard spatially flat results. The ensuing expressions for T and T_G in Eqs. (29)-(30) render the role of spatial curvature explicit and reveal an important structural asymmetry. Although the standard Gauss-Bonnet invariant $\bar{G}$ and its teleparallel counterpart T_G coincide on flat FLRW backgrounds, they differ by curvature-dependent boundary contributions whenever $k \neq 0$.

This inequivalence has direct implications for the dynamics of theories that couple nonlinearly to these invariants. Particularly, it distinguishes teleparallel Gauss–Bonnet gravity from its curvature-based counterpart at the dynamical level.

As an initial application of this framework, we investigated the $-T + F(T_G)$ sector of teleparallel Gauss–Bonnet gravity using a scalar-field representation combined with dynamical-systems techniques. The phase-space structure is richer than in the flat case. The late-time attractor $\mathcal{P}_4^k$ lies on the flat submanifold $\Omega_k = 0$ and trajectories within its basin of attraction asymptotically suppress the curvature contribution. The saddle points $\mathcal{P}_5^k$ and $\mathcal{P}_6^k$ characterized by $q = -1$, correspond to transient de Sitter-like phases within phase space. The curved attractor line $\mathcal{P}_1^k$ demonstrates that spatial flatness is not a universal dynamical outcome. Trajectories lying in its basin asymptotically approach states with non-vanishing Ω_k with the sign of the curvature contribution preserved. The coexistence of flat and curved attractors shows that spatial curvature qualitatively alters the late-time dynamics.

The tetrad construction is not specific to the $\mathcal{F}(T, T_G)$ theory. The irreducible torsion components derived in Appendix B can be used in a broader class of torsion-based theories. The tetrad construction and the resulting torsion invariants are independent of the matter sector. Thus, the framework can be extended to cosmological models containing perfect fluids or minimally coupled scalar fields.

An important next step is to examine the implications of the present construction at the level of cosmological perturbations, where differences between teleparallel and curvature-based formulations are expected to become more pronounced. Such an analysis would simultaneously serve as a perturbative consistency test for the tetrad, since differently rotated tetrads for closed cosmology are known to produce overconstrained perturbative systems [28]. Extending the dynamical systems analysis to include radiation and matter components would allow a more complete cosmological history to be mapped out, including the transitions between matter-dominated and accelerated epochs in the presence of non-zero curvature. Applying the unified tetrad to other modified teleparallel theories would help determine whether the curvature-suppressing behavior is generic to torsion-based gravity or specific to the Gauss–Bonnet sector.

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Appendix A: Field Equations of $F(T, T_G)$ gravity

The field equation after variation of action (10), with respect to tetrads $e^r{}_\rho$ is given by

$$\begin{aligned} \mathbb{E}_r{}^\rho := & \frac{1}{e} \mathcal{F}_T \partial_\sigma (e S_r{}^{\rho\sigma}) + S_r{}^{\rho\sigma} (\partial_\sigma \mathcal{F}_T) + \mathcal{F}_T T^s{}_{\sigma r} S_s{}^{\rho\sigma} + \frac{1}{4} \mathcal{F} e_r{}^\rho - \frac{1}{2} \mathcal{F}_{T_G} \delta_{mnop}^{qstu} e_u{}^\rho \mathcal{K}^{mn}{}_q \mathcal{K}^o{}_{es} \partial_r \mathcal{K}^{ep}{}_t \\ & + \frac{1}{4e} \partial_\sigma \left[\eta_{rp} (\mathcal{A}^{s[ph]} - \mathcal{A}^{h[ps]} + \mathcal{A}^{p[sh]}) e_h{}^\sigma e_s{}^\rho \right] + \frac{1}{4e} T_{mrs} e_h{}^\rho (\mathcal{A}^{s[mh]} - \mathcal{A}^{h[ms]} + \mathcal{A}^{m[sh]}) = 0, \end{aligned} \quad (\text{A1a})$$

with definitions

$$\mathcal{A}^r{}_{mn} = e \mathcal{F}_{T_G} \mathcal{B}^r{}_{mn} - 2 \delta_{vmop}^{tsru} \partial_\rho (e \mathcal{F}_{T_G} e_u{}^\rho \mathcal{K}^{vp}{}_t \mathcal{K}^o{}_{ms}), \quad (\text{A1b})$$

$$\begin{aligned} \mathcal{B}^r{}_{mn} := & \frac{\partial T_G}{\partial \mathcal{K}^r{}_{mn}} \\ = & \mathcal{K}_n{}^e{}_s \mathcal{K}^o{}_{ft} \mathcal{K}^{fp}{}_u \delta_{meop}^{rstu} + \mathcal{K}_m{}^e{}_s \mathcal{K}^o{}_{ft} \mathcal{K}^{fp}{}_u \delta_{enop}^{rstu} + \mathcal{K}^o{}_{et} \mathcal{K}^{ef}{}_s \mathcal{K}_n{}^p{}_u \delta_{omfp}^{tsru} + \mathcal{K}^f{}_{eu} \mathcal{K}^{ep}{}_s \mathcal{K}^o{}_{mt} \delta_{fpon}^{ut sr} \\ & + 2 \mathcal{K}^o{}_{es} \mathcal{K}^{ep}{}_f \mathcal{K}^f{}_{tu} \delta_{mnop}^{rstu} - 2 \mathcal{K}^o{}_{es} \mathcal{K}^e{}_{ft} \mathcal{K}^{fp}{}_u \delta_{mnop}^{rstu} - 2 \mathcal{K}^{oe}{}_s \mathcal{K}_n{}^p{}_ft \mathcal{K}^{fp}{}_u \delta_{omnp}^{srtu} - 2 \mathcal{K}^{ef}{}_t \mathcal{K}^o{}_{ms} \mathcal{K}_n{}^p{}_u \delta_{efop}^{tsru} \\ & - 2 \mathcal{K}^{fp}{}_u \mathcal{K}^o{}_{es} \mathcal{K}^e{}_{mt} \delta_{fpon}^{ut sr} + 2 \mathcal{K}^{oe}{}_s \mathcal{K}_n{}^p{}_f \mathcal{K}^f{}_{tu} \delta_{omnp}^{srtu} + 2 \mathcal{K}^o{}_{es} \partial_u \mathcal{K}^{ep}{}_t \delta_{mnop}^{rstu} + 2 \mathcal{K}^{oe}{}_s \partial_u \mathcal{K}_n{}^p{}_t \delta_{omnp}^{srtu} \\ & + 2 \mathcal{K}^{ep}{}_f \mathcal{K}^o{}_{ms} \mathcal{K}^r{}_{tu} \delta_{epon}^{ftu} + 2 \mathcal{K}^{ft}{}_u \mathcal{K}^u{}_{es} \mathcal{K}^{ep}{}_m \eta_{qn} \delta_{ftop}^{utqr}, \end{aligned} \quad (\text{A1c})$$

and

$$S_r{}^{\rho\sigma} = \frac{1}{2} (\mathcal{K}^{\rho\sigma}{}_r - \delta_r{}^\rho T_\gamma{}^{\gamma\sigma} + \delta_r{}^\sigma T_\gamma{}^{\gamma\rho}). \quad (\text{A1d})$$

Here, all Latin indices are Lorentz or local indices.

Appendix B: Torsion Tensor Irreducible Components

In teleparallel geometry, the torsion tensor can be partitioned into three fundamental irreducible constituents [50]

$$a_\rho := \frac{1}{6} \epsilon_{\rho\sigma\tau\mu} T^{\sigma\tau\mu}, \quad v_\rho := T^\tau{}_\tau{}_\rho, \quad t_{\tau\rho\sigma} := \frac{1}{2} (T_{\tau\rho\sigma} + T_{\rho\tau\sigma}) + \frac{1}{6} (g_{\sigma\tau} v_\rho + g_{\sigma\rho} v_\tau) - \frac{1}{3} g_{\tau\rho} v_\sigma, \quad (\text{B1})$$

which characterize the axial, vector, and tensor sectors respectively. Here, $\epsilon_{\rho\sigma\tau\mu}$ denotes the totally antisymmetric Levi-Civita tensor in four dimensions. On the basis of this mode decomposition, it is then possible to construct several independent scalar curvature invariants associated with the gravitational field

$$T_{\text{ax}} := a_\rho a^\rho, \quad T_{\text{vec}} := v_\rho v^\rho, \quad T_{\text{ten}} := t_{\tau\rho\sigma} t^{\tau\rho\sigma}. \quad (\text{B2})$$

These quantities represent the complete set of non-parity-violating scalar invariants arising from quadratic contractions of the torsion tensor. The standard torsion scalar T is recovered through a specific linear combination of these axial, vector, and purely tensorial invariants [13]

$$T := \frac{3}{2} T_{\text{ax}} + \frac{2}{3} T_{\text{ten}} - \frac{2}{3} T_{\text{vec}}. \quad (\text{B3})$$

For the generic tetrad (22), the irreducible components (B2) are given by

$$a_\rho = \left(-\frac{2N\kappa}{a}, 0, 0, 0 \right), \quad (\text{B4})$$

$$v_\rho = \left(-\frac{3\dot{a}}{a}, 0, 0, 0 \right), \quad (\text{B5})$$

$$t_{\tau\rho\sigma} = 0 \quad \forall \tau, \rho, \sigma \in \{0, 1, 2, 3\}. \quad (\text{B6})$$

and the scalar invariants are found to be

$$T_{\text{ax}} = -\frac{4\kappa^2}{a^2} = -\frac{4k}{a^2}, \quad T_{\text{vec}} = -\frac{9\dot{a}^2}{a^2 N^2}, \quad T_{\text{ten}} = 0. \quad (\text{B7})$$

which reproduce (29).

Appendix C: Stability Analysis Solutions

$$\delta\mathcal{X}(\Sigma) = e^{-4\Sigma} C_1 - \frac{2(1 + \Omega_{k0})}{\Omega_{k0} - 1} (e^{-2\Sigma} - e^{-4\Sigma}) C_2,$$

$$\delta\mathcal{Z}(\Sigma) = e^{-2\Sigma} C_2, \quad (\text{C1})$$

$$\delta\Omega_k(\Sigma) = C_3 + 2\Omega_{k0} (1 - e^{-2\Sigma}) C_2 - \frac{\lambda\Omega_{k0}}{8(\Omega_{k0} - 1)} (1 - e^{-4\Sigma}) [(\Omega_{k0} - 1)C_1 + 2(1 + \Omega_{k0})C_2].$$

Where C_1, C_2 and C_3 are constants of integration determined by initial perturbation amplitudes.

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